

Planet of The Aves: An Immersive Educational Mobile Application for Learning Bird Habitats Through Interactive 3D Visualization and Multimedia Learning

1st Kelvin Nabil Anshary

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
kelvin.anshary@binus.ac.id

2nd Deas Putra Fajar Ramadhan

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
deas.ramadhan@binus.ac.id

3rd Zahra Pinggala Khairunnisa

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
zahra.pinggala@gmail.com

4th Filemon Matthew Budijanto

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
filemon.budijanto@binus.ac.id

5th Felix Christian Tedjawidjaja

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
felix.tedjawidjaja@binus.ac.id

6th Greito Bintang Fulisesa

Computer Science, School of Computer Science

Bina Nusantara University
Bandung Campus, Jakarta, Indonesia
greito.fulisesa@binus.ac.id

Abstract—The advancement of digital technology has transformed educational methods by enabling more interactive and engaging learning experiences. However, learning materials about bird habitats, adaptation, and species characteristics are still commonly delivered through text-based explanations and static images, which may reduce user engagement and limit conceptual understanding. To address this issue, this project proposes Planet of The Aves, an immersive educational mobile application designed to support bird-related learning through interactive multimedia content.

The application was developed using the Agile Software Development Life Cycle (SDLC) methodology and implemented using Flutter. Several educational and immersive features were integrated into the system, including habitat exploration, bird information pages, species distribution maps, interactive 3D bird visualization, habitat ambience audio, and quiz-based assessments. These features aim to provide a more engaging learning experience while improving users' understanding of bird habitats and adaptation characteristics.

The resulting application successfully integrates multimedia learning elements and interactive exploration features into a single mobile platform. Future development will focus on the integration of Augmented Reality (AR) technology to support a more immersive mixed reality learning experience.

Keywords—Immersive Learning, Educational Application, Mobile Application, Flutter, Interactive Multimedia, 3D Visualization, Bird Habitat Learning, Augmented Reality.

I. INTRODUCTION

A. Background

The rapid development of digital technology has significantly influenced the education sector by providing new

approaches to learning and knowledge delivery [1]. Traditional learning methods often rely on textbooks, written explanations, and static images, which may not fully engage learners, particularly when studying subjects that require visualization and exploration [2]. As a result, educators are increasingly adopting multimedia-based learning approaches to improve student engagement and understanding [3].

One of the topics in biology that can benefit from interactive learning is the study of aves or birds. Birds exhibit diverse habitat characteristics, adaptation mechanisms, and distribution patterns depending on their environmental conditions. Different bird species can be found in tropical forests, wetlands, savannahs, and arctic regions, each possessing unique physical and behavioral adaptations that enable survival within their respective ecosystems [4].

Despite the richness of this subject, educational materials about birds are still commonly presented through text-based explanations and two-dimensional images. Such approaches may limit learners' ability to visualize environmental conditions and understand the relationship between birds and their habitats [5]. Consequently, users may experience reduced motivation and lower levels of engagement during the learning process.

Immersive learning technologies provide opportunities to address these limitations by integrating multimedia elements, interactive visualization, and user-centered exploration into educational systems [6]. Interactive learning environments have been shown to increase learner engagement, motivation, and knowledge retention by allowing users to actively participate in the learning experience [7].

Based on these considerations, this project proposes Planet of The Aves, an immersive educational mobile application designed to support bird habitat learning through interactive

multimedia features. The application integrates habitat exploration, bird information pages, distribution maps, interactive 3D bird visualization, habitat ambience audio, and quiz-based assessments to create a more engaging educational experience.

B. Problem Statement

Although information regarding bird habitats and adaptation is widely available, several challenges remain in current learning approaches:

1. Educational materials about birds are still predominantly delivered through text and static images.
2. Users often experience difficulties in visualizing bird habitats and understanding adaptation characteristics through conventional learning methods.
3. Existing educational media generally provide limited interactivity, reducing user engagement during the learning process.
4. Mobile educational applications that combine multimedia learning, 3D visualization, and immersive interaction for bird habitat education remain limited.

C. Proposed Solutions

To address the identified problems, this project proposes the development of Planet of The Aves, an immersive educational mobile application focused on bird habitat learning.

The application provides several interactive features, including habitat selection, bird information pages, species distribution maps, interactive 3D bird visualization, habitat ambience audio, and educational quizzes. These features are integrated into a single platform to support a more engaging and effective learning experience.

In addition, future development of the application will explore the integration of Augmented Reality (AR) technology to provide a mixed reality learning environment where users can visualize bird models within real-world surroundings.

D. Objective

The objectives of this project are as follows:

1. To develop an immersive educational mobile application for learning bird habitats and adaptation characteristics.
2. To provide interactive learning experiences through multimedia content and 3D bird visualization.
3. To improve user engagement and interest in biology-related learning materials.
4. To support users in understanding bird habitats, distribution patterns, and environmental adaptation more effectively.
5. To establish a foundation for future integration of Augmented Reality (AR) technology in educational bird-learning applications.

E. Scope of Project

Although information regarding bird habitats and adaptation is widely available, several challenges remain in current learning approaches:

This project focuses on the development of an educational mobile application that introduces selected bird species from four habitat categories: Tropical Forest, Wetland, Savannah, and Arctic Area.

The application includes the following features:

1. Habitat exploration.
2. Bird information pages.
3. Bird distribution maps.
4. Interactive 3D bird visualization.
5. Habitat ambience audio.
6. Educational quizzes and automatic score evaluation.

The application is intended for educational purposes and targets students as well as general users who are interested in learning about birds, habitats, and biological adaptation concepts.

II. LITERATURE REVIEW

A. Immersive Learning

Immersive learning is an educational approach that utilizes digital technology and interactive media to create engaging learning experiences [8]. Unlike conventional learning methods, immersive learning encourages active participation by allowing users to interact with educational content through visualization, exploration, and multimedia elements. Research has shown that immersive learning environments can improve learner motivation, engagement, and knowledge retention by providing more realistic and meaningful learning experiences [9].

Various technologies have been applied to immersive learning systems, including virtual reality (VR), augmented reality (AR), interactive multimedia, and 3D visualization. These technologies help learners understand complex concepts by transforming abstract information into interactive visual experiences [10].

In this project, immersive learning principles are implemented through habitat exploration, interactive 3D bird visualization, habitat ambience audio, and quiz-based assessments to create a more engaging educational experience.

B. Educational Mobile Application

Planet of The Aves is categorized as an educational application because it provides educational content related to bird habitats, adaptation, and reproduction lifecycle through immersive and interactive learning features.

Educational mobile applications are software systems designed to facilitate learning through portable digital devices such as smartphones and tablets [11]. Mobile learning applications provide flexibility by allowing users to access educational content anytime and anywhere. In addition, mobile applications can integrate multimedia elements such as images, audio, animation, and interactive activities to improve learning effectiveness.

Recent studies indicate that educational mobile applications can increase learner engagement and encourage self-directed learning by providing interactive and accessible educational resources [12]. As a result, mobile platforms have become one of the most popular environments for delivering modern educational content.

Planet of The Aves adopts the educational mobile application approach by integrating multimedia-based learning materials into a smartphone-based platform.

C. Interactive Multimedia Learning

Interactive multimedia learning combines multiple forms of media, including text, graphics, audio, animation, video, and user interaction, into a single educational environment [13]. According to multimedia learning theory, combining visual and auditory information can improve users' understanding and memory retention compared to traditional text-only learning methods [14].

Interactive multimedia systems also encourage active learning by allowing users to explore educational content through navigation, interaction, and feedback mechanisms. Features such as quizzes, interactive visualizations, and multimedia presentations have been widely used to improve educational outcomes [15].

In this project, interactive multimedia learning is implemented through bird information pages, habitat ambience audio, distribution maps, interactive 3D models, and educational quizzes.

D. Aves Habitat and Adaptation

Aves, commonly known as birds, are warm-blooded vertebrates that occupy a wide variety of habitats across the world [16]. Different bird species exhibit unique physical and behavioral adaptations that enable survival in specific environmental conditions. Habitat characteristics such as temperature, vegetation, food availability, and climate significantly influence bird adaptation and distribution patterns.

For example, birds inhabiting tropical forests often possess strong flight capabilities and vibrant coloration, while birds living in arctic environments develop insulating feathers to withstand extremely low temperatures [17]. Understanding these habitat-adaptation relationships is an important component of biological education because it demonstrates how organisms interact with their ecosystems.

This project utilizes selected bird species from Tropical Forest, Wetland, Savannah, and Arctic habitats as educational content to support learning about bird adaptation and environmental diversity.

E. Agile Software Development Life Cycle

The Software Development Life Cycle (SDLC) provides a structured framework for software development activities [18]. One of the most widely adopted SDLC methodologies is Agile, which emphasizes iterative development, collaboration, flexibility, and continuous improvement throughout the development process [19].

Agile development allows software teams to implement features incrementally while continuously evaluating and improving system quality. This methodology is particularly suitable for projects involving user interface design, interactive features, and evolving requirements [20].

In this project, Agile SDLC was selected because it supports incremental feature development and allows continuous refinement of immersive educational components throughout the development process.

F. Interactive 3D Visualization

In this project, interactive 3D exploration is implemented to allow users to explore bird models, habitats, and lifecycle objects interactively within the application.

Interactive 3D visualization enables users to observe and interact with three-dimensional digital objects through rotation, zooming, and exploration features [21]. Compared to static images, 3D visualization provides richer spatial information and allows users to understand object structures more effectively.

Educational applications frequently utilize 3D visualization to improve concept comprehension and increase learner engagement. Previous studies have demonstrated that interactive 3D models can support experiential learning and enhance users' ability to explore educational content [22].

In Planet of The Aves, interactive 3D visualization is implemented using GLB-based bird models, allowing users to observe bird appearances from multiple perspectives through direct interaction.

G. Augmented Reality in Education

Augmented Reality (AR) is a technology that combines virtual objects with real-world environments in real time [23]. AR has gained significant attention in education because it can provide immersive learning experiences while maintaining user awareness of the surrounding environment.

Educational AR applications have been used in science, biology, engineering, and medical education to improve visualization and learner engagement [24]. By integrating virtual objects into real-world settings, AR can help users understand abstract concepts more effectively and increase motivation during learning activities.

Although AR functionality has not yet been implemented in the current version of Planet of The Aves, it is considered a promising future enhancement for supporting mixed reality learning experiences involving interactive bird visualization.

III. METHODS

A. SDLC Method

This project adopts the Agile Software Development Life Cycle (SDLC) methodology. Agile was selected because it supports iterative and incremental software development while allowing continuous evaluation and improvement throughout the development process [19].

The Agile approach enables the development team to implement application features gradually, test individual components, and refine the system based on project requirements. This methodology is suitable for Planet of The Aves because the application contains multiple interactive and multimedia-based features that require continuous adjustments during development.

The development process was carried out incrementally, beginning with the application interface and navigation system, followed by habitat modules, bird information pages, quiz implementation, multimedia integration, and 3D visualization features.

B. Planning

The planning stage focused on defining the educational objectives, target users, application scope, and main system features.

Planet of The Aves was designed as an immersive educational mobile application intended to improve users' understanding of bird habitats and adaptation characteristics. The application targets students and general users interested in biology and environmental education.

Four habitat categories were selected as learning modules:

1. Tropical Forest
2. Wetland
3. Savannah
4. Arctic Area

Several bird species were chosen from each habitat to represent different environmental conditions and adaptation mechanisms.

At this stage, the development team also established the application structure, navigation flow, content requirements, and development responsibilities.

C. Requirement Analysis

Requirement analysis was conducted to identify the functional and non-functional requirements of the application.

1. Functional Requirements

The system shall:

- Allow users to select a habitat environment.
- Allow users to access bird information pages.
- Display bird distribution maps.
- Provide interactive 3D bird visualization.
- Play habitat ambience audio.
- Provide educational quizzes.
- Calculate and display quiz scores automatically.
- Allow users to navigate between modules and learning pages.

2. Non-Functional Requirements

The application shall:

- Provide responsive navigation.
- Deliver a user-friendly interface.
- Support Android mobile devices.
- Maintain stable multimedia playback.
- Provide smooth interaction with 3D visualization components.
- Ensure maintainability and scalability for future feature development.

D. System Design

The system design stage focused on designing application workflows, navigation structures, and user interfaces.

The application flow begins from the Welcome Screen, followed by Habitat Selection, Bird Selection, Bird Information Page, 3D Visualization Page, Quiz Module, and Quiz Result Page.

The design process was conducted using Figma to create interface prototypes before implementation. User interaction was designed to remain simple and intuitive to support educational learning activities.

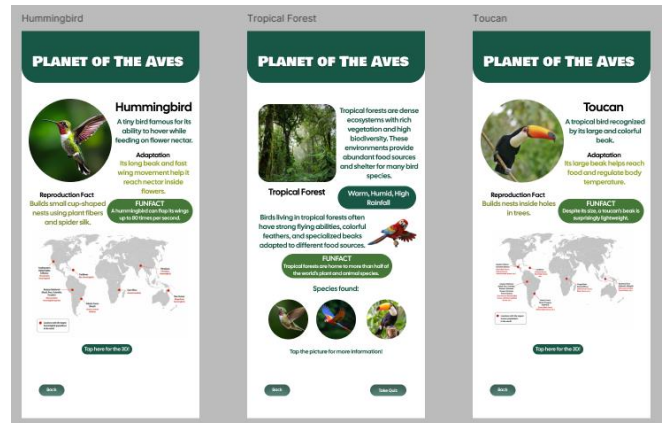


Figure 1 presents the Main User Interface Prototype designed in Figma

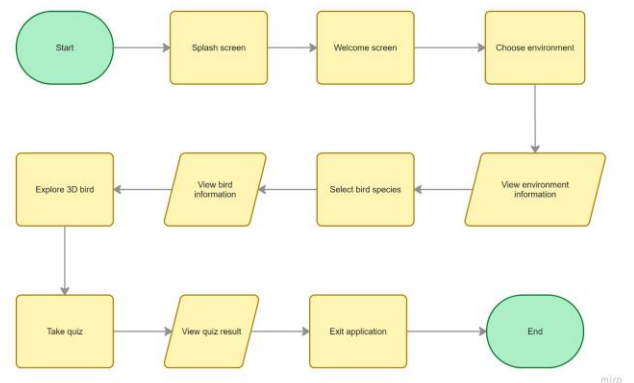


Figure 2 presents the Application Flowchart.

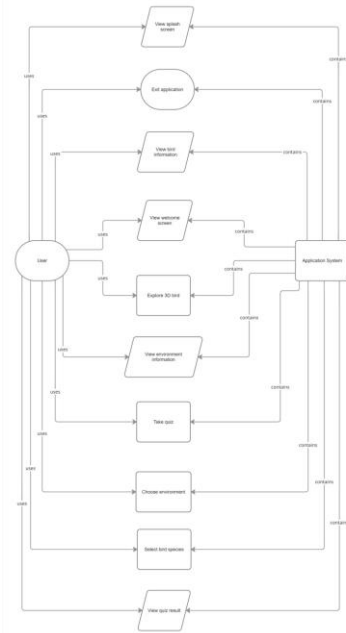


Figure 3 presents the Use Case Diagram of the system.

E. Asset Preparation

Several multimedia assets were prepared to support the immersive learning experience within the application.

The assets include:

1. Bird images.
2. Bird distribution maps.
3. GLB-based 3D bird models.
4. Habitat ambience audio files.

5. Habitat illustrations.
6. Application icons and interface assets.

These resources were integrated into the application to provide visual and auditory learning experiences that improve user engagement and understanding.

F. Application Development

The implementation stage focused on translating the system design into a functional mobile application.

Planet of The Aves was developed using Flutter as the primary framework and Dart as the programming language. Flutter was selected because it provides cross-platform mobile development capabilities and supports modern user interface design.

Several supporting technologies were utilized during development:

1. Flutter Framework
2. Dart Programming Language
3. Android Studio
4. Visual Studio Code
5. GitHub
6. Figma
7. model_viewer_plus
8. audio players

The application architecture follows a widget-based approach utilizing StatelessWidget and StatefulWidget components to promote modularity and code maintainability.

The implemented modules include:

1. Welcome Screen
2. Habitat Selection Module
3. Bird Information Module
4. Distribution Map Module
5. Interactive 3D Visualization Module
6. Habitat Audio Module
7. Quiz Module
8. Quiz Result Module

Future development is planned to include Augmented Reality (AR) integration, enabling users to visualize bird models within real-world environments through mobile devices.

G. Testing and Improvement

Testing was performed using the Black Box Testing approach to verify application functionality.

The testing process focused on evaluating:

- Navigation functionality.
- Bird information display.
- Distribution map display.
- 3D visualization functionality.
- Habitat audio playback.
- Quiz functionality.
- Quiz score calculation.

After testing was completed, several improvements and bug fixes were performed to improve application stability, multimedia performance, and user experience.

The iterative nature of Agile development enabled continuous refinement of features throughout the

implementation process, resulting in a more stable and functional educational application.

IV. EXPERIMENTAL RESULTS

A. Application Implementation

The final version of Planet of The Aves successfully implements several educational and immersive learning features.

1. Welcome Screen and Habitat Selection

The application begins with a welcome screen followed by a habitat selection page. Users can choose one of four available habitats: Tropical Forest, Wetland, Savannah, and Arctic Area.

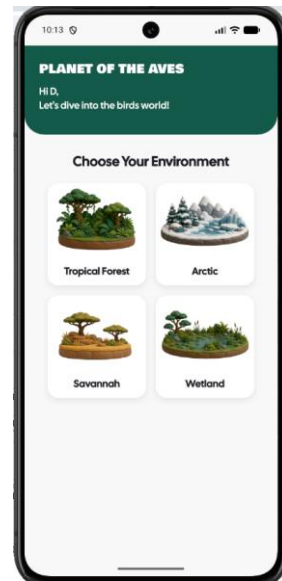


Figure 4 shows the Habitat Selection interface.

2. Bird Information Module

Each habitat contains several bird species that users can explore. The bird information page provides educational content regarding bird characteristics, adaptation, habitat descriptions, and interesting facts.



Figure 5 presents an example of the Bird Information Page.

3. Bird Distribution Maps

To improve geographical understanding, each bird species is accompanied by a distribution map illustrating its habitat range and natural distribution area.

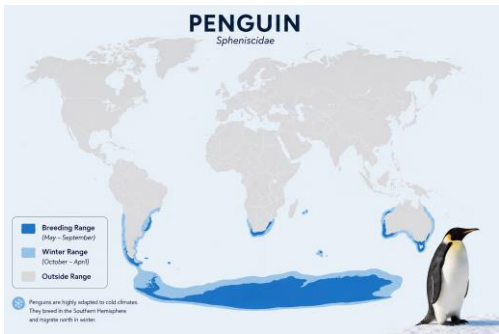


Figure 6 presents an example of a Bird Distribution Map.

4. Interactive 3D Bird Visualization

One of the main immersive features of the application is the integration of interactive 3D bird models. Users can access a dedicated 3D visualization page and explore bird models from different viewing angles.

The implementation utilizes GLB-based 3D models rendered through the model_viewer_plus package.



Figure 7 presents the Interactive 3D Bird Visualization Module.

5. Habitat Ambience Audio

To increase immersion, environmental ambience audio is integrated into each habitat category. Different habitat environments are represented by different audio experiences corresponding to the selected ecosystem.

This feature aims to provide a richer educational atmosphere and improve user engagement.

6. Quiz Module

The application includes an educational quiz module consisting of multiple-choice questions related to bird habitats and adaptation concepts.

After completing the quiz, users receive automatic scoring and performance feedback.

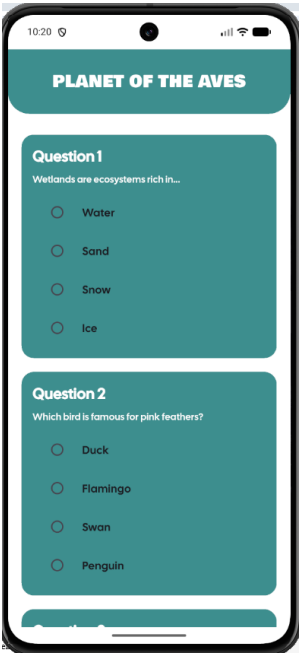


Figure 8 presents the Quiz Interface.



Figure 9 presents the Quiz Result Page.

B. Testing Result

Black Box Testing was conducted to evaluate application functionality based on user interactions and expected outputs. Table I presents the testing results.

No	Feature	Expected Result	Status
1	Habitat Selection	Habitat displayed correctly	Passed
2	Bird Information	Information displayed correctly	Passed
3	Distribution Map	Map displayed correctly	Passed

4	Habitat Audio	Audio played successfully	Passed
5	Quiz System	Questions displayed correctly	Passed
6	Score Calculation	Score calculated automatically	Passed
7	Navigation System	Navigation functioned properly	Passed
8	3D Visualization	3D Model displayed successfully	Passed

The testing results indicate that all primary features operated successfully according to their intended functionality. The application demonstrated stable performance during both emulator testing and physical Android device testing.

C. Discussion

The implementation results demonstrate that immersive multimedia elements can be successfully integrated into a mobile educational application. The combination of habitat exploration, bird information, distribution maps, 3D visualization, ambience audio, and quizzes provides users with a more engaging learning experience compared to traditional text-based learning materials.

The integration of interactive 3D visualization and environmental audio contributes significantly to the immersive characteristics of the application. Furthermore, the modular architecture developed using Flutter allows future feature expansion, including the planned integration of Augmented Reality (AR) technology.

Overall, Planet of The Aves successfully fulfills its primary objective of providing an interactive and immersive platform for learning bird habitats and adaptation characteristics.

V. CONCLUSION

This project successfully developed Planet of The Aves, an immersive educational mobile application designed to support learning about bird habitats, adaptation characteristics, and species distribution through interactive multimedia technologies.

The application was developed using the Agile Software Development Life Cycle (SDLC) methodology and implemented using Flutter. Several educational features were successfully integrated, including habitat exploration, bird information pages, bird distribution maps, interactive 3D bird visualization, habitat ambience audio, and quiz-based assessments.

The implementation results demonstrate that multimedia learning components can enhance user engagement and provide a more interactive educational experience compared to traditional text-based learning methods. Furthermore, the integration of 3D visualization and environmental audio contributes to the immersive nature of the learning environment.

Based on functional testing results, all major features operated successfully according to their intended objectives. The application was also successfully built as an Android application package (APK) and tested on Android devices.

For future development, Augmented Reality (AR) technology can be integrated to enable users to visualize bird models directly within real-world environments. Additional improvements may include animated bird models, expanded species collections, cloud-based learning progress tracking, and more advanced educational interactions.

The results of this project indicate that immersive multimedia learning applications have strong potential to support biology education and improve user engagement in environmental and wildlife learning topics.

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CONTRIBUTION

Kelvin Nabil Anshary contributed to project planning, requirement analysis, software engineering documentation, project coordination, and report preparation.

Deas Putra Fajar Ramadhan contributed to application development, Flutter implementation, multimedia integration, 3D visualization implementation, habitat audio integration, testing, and deployment.

Zahra Pinggala Khairunnisa contributed to UI/UX design, educational content preparation, interface evaluation, and documentation support.

Filemon Matthew Budijanto contributed to project research, literature review preparation, reference collection, and testing activities.

Felix Christian Tedjawidjaja contributed to software design evaluation, feature validation, system testing, and quality assurance activities.

Greito Bintang Fulisesa contributed to data collection, educational content verification, application evaluation, and project documentation support.

OPEN DATA ACCESS

The source code, project documentation, and application resources used in this study are maintained in the project repository. The repository link will be provided as part of the final project submission through GitHub.

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